Evolving Multi-Segmented Creatures for Mountain Climbing

CM3020 Artificial Intelligence — Mid-Term Coursework (Part B)

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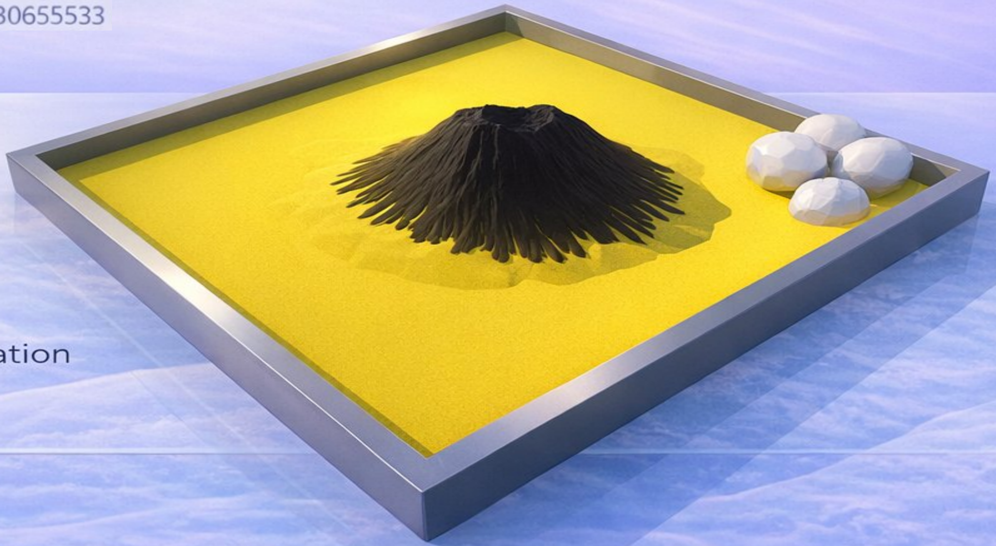
Environment:

25x25 Sandbox with central Gaussian Mountain

Experiment Domain:

Evolutionary Robotics in Physics Simulation (PyBullet)



Research Question & Experimental Philosophy

Why does evolution fail before it succeeds in physics-based locomotion tasks?

Focus:

- Identifying reward exploits
- Eliminating failure modes systematically
- Separating search limitations from physical constraints

Key Principle:

Each experiment isolates **one failure source.**

Baseline Exploits (Experiments 1–3)

Baseline Objective

- Establish how the evolutionary system behaves under a naive, height-driven fitness function.

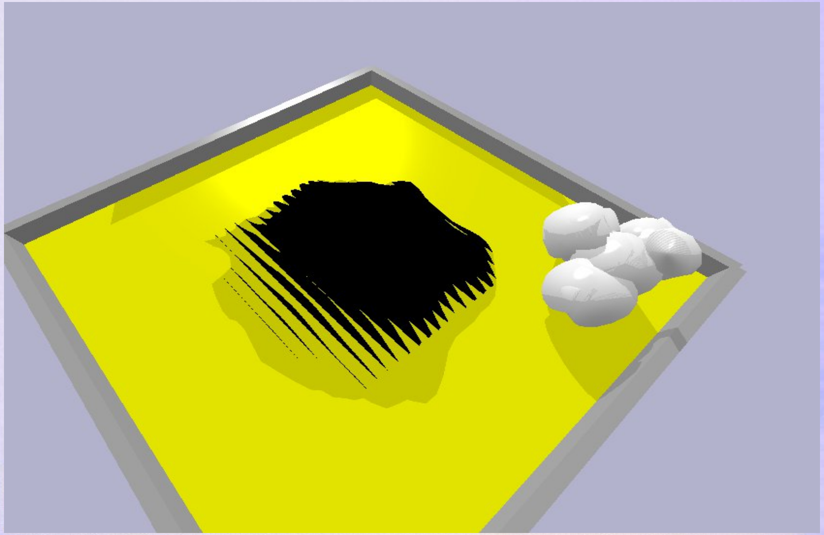
Fitness Definition

- Final Height × Contact Multiplier

(No directional, posture, or size constraints)

Observed Behaviour

- Creatures increased overall body size to inflate achievable height
- Individuals leaned against arena walls or wedged into corners for passive support
- No coordinated locomotion or forward movement emerged



Wall-supported height maximisation and body size inflation under baseline fitness

Key Insight

- The fitness function rewarded static height maximisation, not purposeful

Evolution optimised the metric through morphological scaling and environmental support rather than solving the locomotion task.

Distance Bonus Failure (Experiment 4)

Addressing Experiment 1–3 Limitation:
Baseline fitness rewarded height without movement.

Purpose

- Introduce directional pressure to encourage movement toward the mountain centre.

Fitness Modification

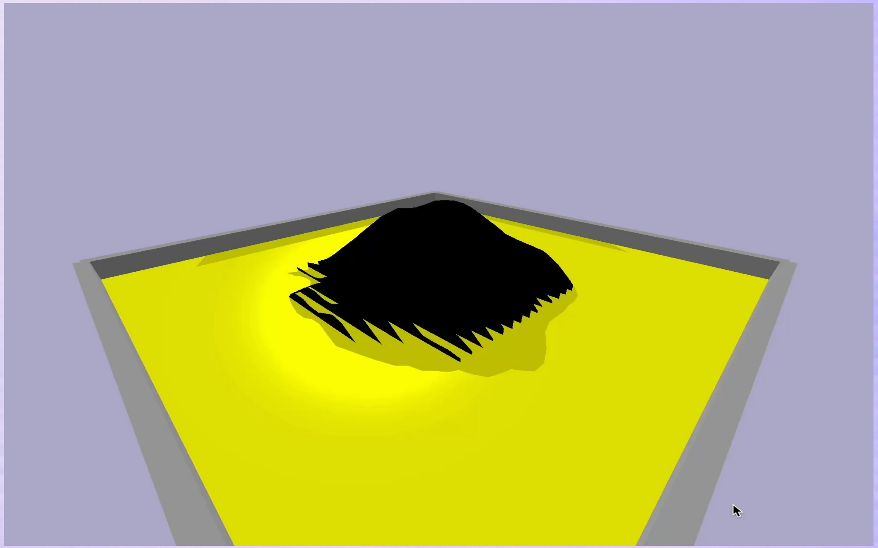
- Height × Distance-to-Centre Bonus (normalised by body size)

Observed Behaviour

- Creatures immediately toppled or “timbered” forward at the start of evaluation
- Large distance reward obtained in a single motion
- No sustained movement, gait formation, or control strategy emerged

Key Insight

- Rewarding displacement without enforcing control intent produces single-action exploitation rather than learned locomotion.



Instant distance exploitation under directional fitness

Experiments 5–6: Upright Constraint → Centipede Breakthrough

Problem (from Experiment 4)

- Directional distance reward caused a “timber” exploit: creatures fell forward to gain distance without control.

Experiment 5: Fitness Constraint Update

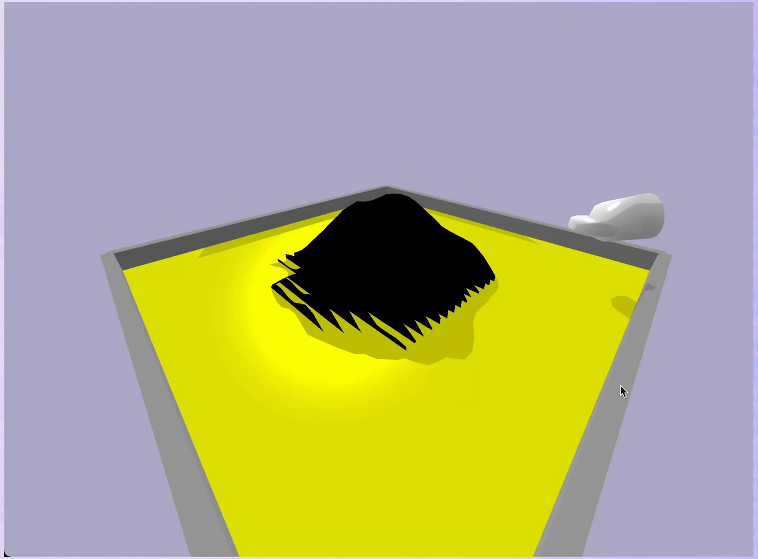
Change Added an upright orientation penalty to the existing height × distance fitness.

Experiment 6: Capacity Increase (Fitness Unchanged)

Change Gene count increased from 3 → 5 (Fitness function unchanged from Experiment 5)

Result:

- Multi-segment coordination emerged
- Stable peristaltic “centipede” gait formed
- Creatures reliably moved from the corner toward the mountain base



First sustained, task-aligned movement without reward exploitation.

Experiment 7: Large-Scale Search Expansion

Hypothesis

Previous failures may be caused by insufficient population size or evolutionary depth.

Baseline Search (Experiments 1–6)

- Population size: **50**
- Generations: **100**
- Fitness function: unchanged from Experiment 6

Expanded Search (Experiment 7)

- Population size increased to **100**
- Generations increased to **300**
- Fitness function unchanged

Result

- No sustained climbing behaviour emerged
- Best-performing individuals still failed to ascend the slope

Conclusion

Failure is **not**: due to limited search.

The bottleneck lies in physical interaction and **morphology**, not optimisation scale.

Advanced Phase — Encoding Redesign & Stability Engineering (Experiment 8)

Why a redesign was necessary

- Reward exploits eliminated (Experiments 1–6)
- Search capacity ruled out (Experiment 7)
- Remaining failures could no longer be attributed to optimisation or noise

Conclusion: A system-level redesign was required.

Advanced Encoding & Simulation Changes

- **Fixed Morphology Encoding**
  - Body size, shape, and joint spacing are no longer evolvable
- **Control-Only Evolution**
  - Evolution operates purely on motor control parameters

Outcome:

A clean, physically meaningful baseline for **hypothesis-driven tests**
(Experiments 9–15).

Stability Engineering

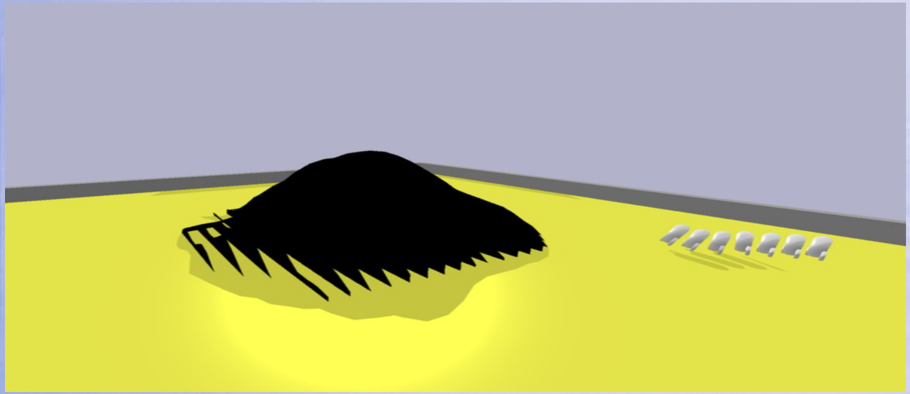
- Passive stabilising legs, coliision filtering, and physics tuning to remove solver artefacts and instability

Why This Is an Advanced Baseline

- Removes morphology-based cheating entirely
- Eliminates simulation explosions and jitter-driven artefacts
- Produces consistent, repeatable behaviour across runs

Experiment 8: Fixed Morphology — Behaviour Under Clean Physics

Approach, contact, and stall without exploits



Observed Behaviour

- The creature reliably reaches the slope
- Contact is maintained without instability
- Upward progress fails due to slipping and poor foothold geometry

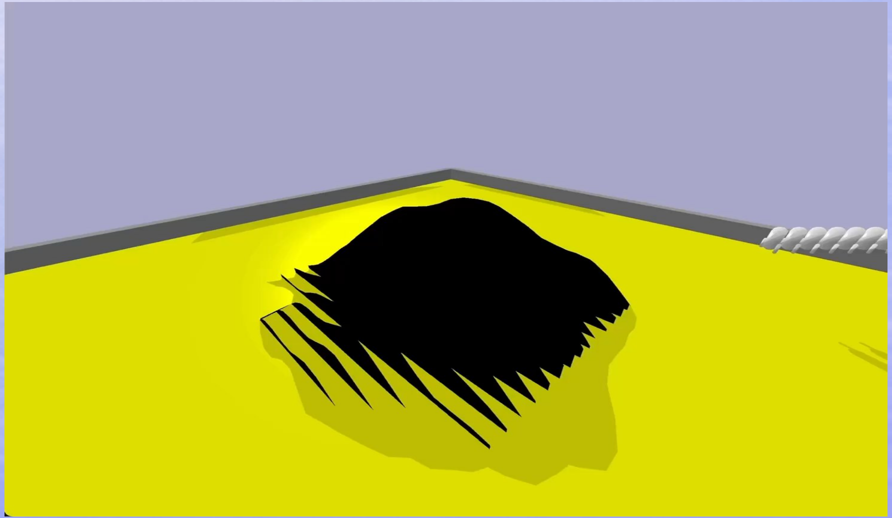
Key Point: This failure occurs **without**:

- Wall exploitation
- Morphology inflation
- Simulation instability

Fixed Encoding Effects

- Body size and shape are clamped
- Joint spacing is analytically fixed
- Passive legs increase contact without adding control DOFs

Result: Morphology-based cheating and physics artefacts are structurally removed.



Physical Isolation Tests: Grip vs Force (Experiments 9–11)

Experiment 9
Mountain Friction Increase
(Grip Hypothesis I)

- Change Applied:
  - Mountain lateral friction increased to 1.8
  - Gene count increased to 7
  - Fixed morphology and fitness unchanged
- Observation:
  - Creature reliably reaches the mountain
  - Improved contact duration and surface interaction
- **No sustained ascent**

Experiment 10
Mountain Friction with Simpler Control (Grip Hypothesis II)

- Change Applied:
- Mountain friction held at 1.8
- Gene count reduced to 5
- Observation:
  - Behaviour qualitatively identical to Experiment 9
- Interpretation:
  - Failure mode invariant to control complexity
  - Bottleneck is not insufficient control resolution

Experiment 11
Motor Force Scaling
(Torque Hypothesis)

- Change Applied:
- Motor force increased from ~20 ~ 70
- High mountain friction retained
- Morphology and fitness unchanged
- Observation:
- Higher impact forces and osciliations
- Increased instability and posture disruption
- **Still no uphill progress**

Conclusion: Neither increased grip nor increased actuation resolves the climbing failure.

Anchoring Test: Floor Friction (Experiments 12–13)

Hypothesis: Insufficient floor traction prevents force transfer intthe slope.

Experiment 12

- Floor lateral friction increased to 2.5
- Population = 50, Generations = 100
- **Findings:**
  - Reduced rear slip and improved grounding
  - Maintained contact and stability at the slope base, but repeated oscillation and sliding prevented any net upward displacement

Experiment 13

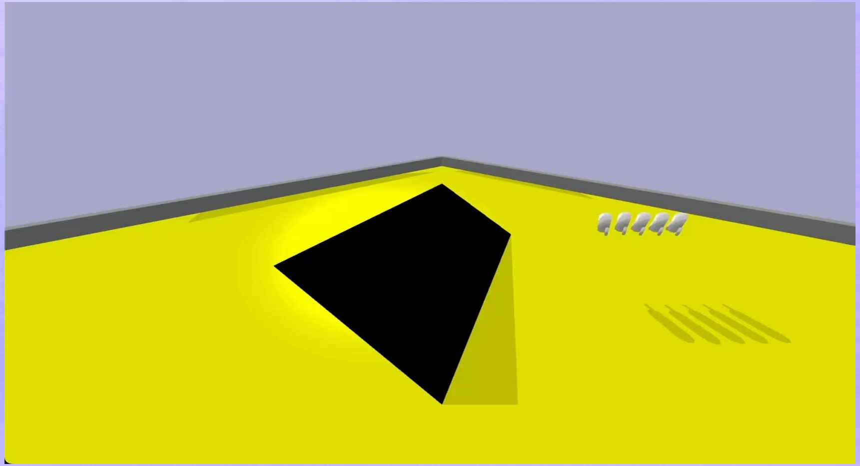
- Floor lateral friction reduced to 1.5
- Population = 50, Generations = 100
- **Findings:**
  - Same qualitative failure pattern observed
  - Confirms anchoring is not the limiting factor

Verdict: Anchoring improves stability but does not unlock climbing.

The bottleneck lies in foothold/contact geometry, **not floor traction**.

Exceptional: Contact-Aware Control & Terrain Validation (Experiments 14–15)

- **Why add contact-gated control?**
  - Prior experiments showed repeated impact → oscillation → slide-back
  - Open-loop motor oscillation wastes force once contact is made
  - A reflex-like response is required to "brace" instead of flail
- **Control architecture change** (Experiments 14–15):
  - Motor actuation conditioned on per-link contact state
  - Contacting links switch to bracing (POSITION_CONTROL)
  - Selected joints anchor with increased force to resist backsliding
- **Why compare terrains?**
  - Gaussian slope tests smooth, idealised contact
  - Irregular mountain URDF introduces realistic footholds and traps
  - Same controller tested under both to validate generality



- Contact-Aware Sensory Feedback
  - Open-loop control caused repeated impacts, oscillation, and back-sliding once the creature contacted the slope.
  - Each body segment detects ground or slope contact
  - Motor output is gated by contact state.
  - Contacting segments switch from oscillation to bracing
  - Non-contacting segments continue propulsion

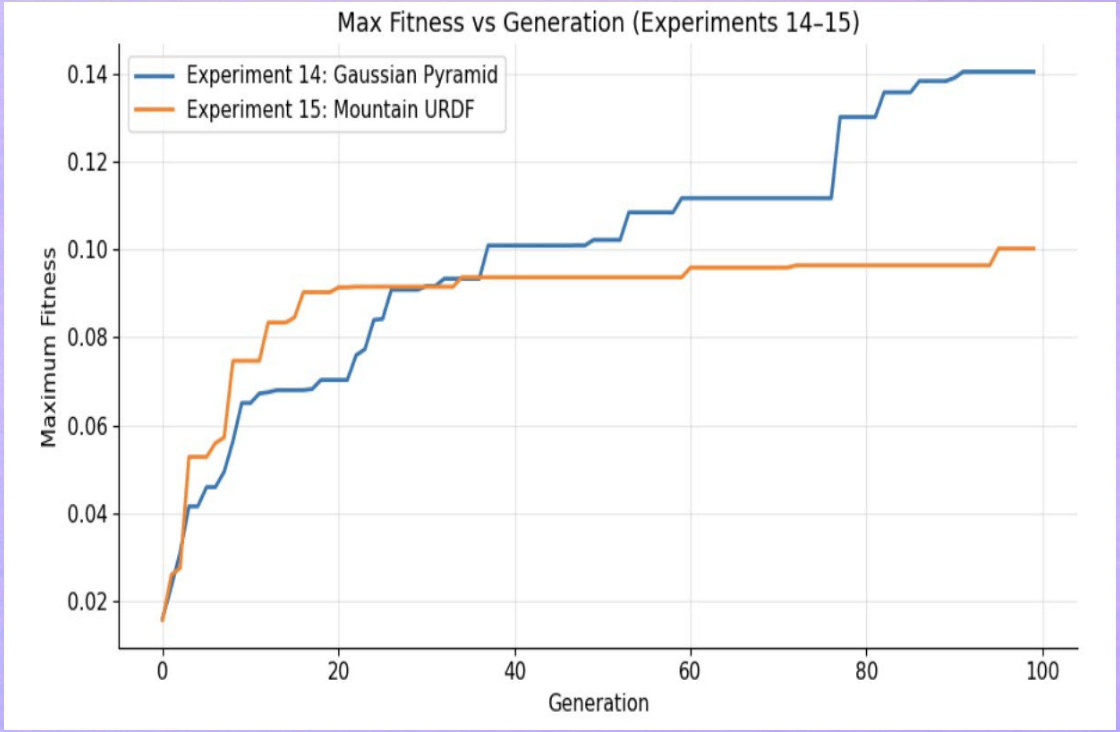
Final validation: Even with contact-aware control and varied terrain, climbing fails due to mechanical interaction geometry, not optimisation, force, or friction.

Exceptional: Fitness Comparison Over Varied Terrain (Experiments 14–15)

- Experiment 14 (Gaussian Pyramid):
  - Produced modest height gains as gradual gradients allowed smoother contact and bracing.
- Experiment 15 (Mountain URDF):
  - Its sharper, irregular terrain prevented foothold stability and caused frequent trapping.

Fitness Comparison Findings:

- Experiment 14 (Gaussian Pyramid) achieved steadily rising fitness, benefiing from the smoother, gradual slope.
- Experiment 15 (Mountain URDF) ptateaued early due to uneven, irregular features disrupting footholds.



Conclusion & Key Findings

Across all experiments, behaviour was limited not by optimisation capacity, but by fundamental physical interaction constraints.

This work demonstrated:

- Reward shaping alone is insufficient for complex locomotion
- Increasing force, friction, or generations does not overcome poor contact geometry
- Sensory feedback improves stability but does not guarantee task success

Key takeaway: Systematic isolation of failure sources is critical to understanding and advancing embodied intelligence.

Evolutionary difficulty is a diagnostic signal, not an optimisation defect.